Kelsey McGuire

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Professional Summary

MSc Candidate in Geography at the University of British Columbia with 3+ years of experience in aquatic and terrestrial environmental sampling. Experienced in environmental field sampling, water quality monitoring, greenhouse gas measurements, statistical analysis (R), GIS, and scientific data visualization. Proven ability to manage field campaigns, analyze environmental datasets, and communicate findings through technical reports.

Technical Skills & Certifications

Programming: R (tidyverse, ggplot2, ShinyApp), Git/GitHub, Python

Software: ArcGIS Pro, QGIS, ENVI, Google Earth Engine, Microsoft Suite

Field Instrumentation: LICOR Trace Gas Analyzers, Hanna/YSI Water Sondes, Soil Coring

Certifications: Remote/Workplace First Aid, Transport Endorsement, Safe Kayaking, Class 5 License, DELF B2

Education

M. Sc., GEOGRAPHY **2024 - 2026**

University of British Columbia (Vancouver, BC)

***Thesis Title**: A little sedge goes a long way: Vegetation as a key driver of littoral Methane flux*

Thesis Advisor(s): Dr. McKenzie Kuhn

GPA: 4.33/4.33 or 93%

B. Sc., GEOGRAPHICAL SCIENCES WITH DISTINCTION **2021 - 2024**

University of British Columbia (Vancouver, BC)

GPA: 3.95/4.33 or 85%; Dean's List Recipient 2021W; 2022W; 2023W

Relevant Work Experience

GRADUATE RESEARCHER| BOREAL-ARCTIC BIOGEOCHEMISTRY LAB, UBC | 09/2025 –

- Collected, processed, quality-controlled, and analyzed water quality and carbon datasets comprising over 300 observations using standardized aquatic sampling techniques, including Van Dorn and grab sampling.
- Coordinated field campaign logistics by procuring equipment, managing project resources, and supporting field operations to ensure efficient data collection.
- Produced technical reports, plain-language knowledge products, and outreach materials for community partners and grant agencies, and presented research findings at two national academic conferences.

UNDERGRADUATE RESEARCH ASSISTANT | GEOSPATIAL ECOLOGY LAB, UBC | 09/2023– 08/2024

- Collected and processed 499 soil samples across six savanna–forest mosaic sites using standardized field and laboratory protocols, including soil coring, sieving, and manual root and charcoal extraction.
- Analyzed and visualized environmental and spatial datasets using R and QGIS to support ecological research and data interpretation.
- Coordinated field campaign logistics by preparing sampling equipment, procuring field supplies, and communicating daily sampling plans with research teams to support efficient field operations.
- Contributed to two meta-analyses by synthesizing scientific literature, extracting and categorizing study variables, and visualizing synthesis results using R.

UNDERGRADUATE RESEARCH ASSISTANT | UBC MICROMETEOROLOGY LAB, UBC | 05/2023 – 03/2024

- Conducted greenhouse gas field measurements at 48 sampling sites within a post-fire peatland ecosystem using standardized field protocols.
- Maintained four eddy covariance towers through routine QA/QC of flux data, calibration and cleaning of LICOR CH_4 and CO_2 analyzers, and maintenance of field instrumentation.
- Developed R scripts to automate PhenoCam vegetation index calculations and created an interactive Shiny application for visualization of eddy covariance carbon flux data.

Relevant Courses

STAT545 | INTRODUCTORY DATA ANALYSIS | UNIVERSITY OF BRITISH COLUMBIA

- Focused on efficiently employing functions within R's tidyverse and ggplot2 packages. Main tasks included learning to properly filter, clean, and present data based on the structure and goals of the given objective.
- Key outputs included a shared GitHub project that employed R Markdowns and displayed the use of many tidyverse functions including filter(), select(), mutate(), alongside a Shiny App that was created to visualize methane data against environmental variables based on a user supplied dataset.

GEOS370 (ADVANCED GEOGRAPHIC INFORMATION SCIENCE); GEOS373 (INTRODUCTORY REMOTE SENSING)

- Completed multiple individual and group projects using geospatial software, such as:
- Used the suitability modeler within ArcGIS Pro, along with Kriging and Nearest Neighbour techniques, to analyse and identify suitable Caribou Habitats in Canada's North that was presented through StoryMaps. Automated portions of this analysis using ArcGIS' built-in python notebook and workflows.
- Used ENVI and Google Earth Engine to conduct numerous analyses on satellite imagery, where collection, pre- and post-processing analyses were performed to compare the effectiveness of supervised versus unsupervised classifiers in identifying sea level rise surrounding New Orleans, LA.

Scholarships & Awards

CGU BEST STUDENT POSTER – BIOGEOSCIENCES	**MAY 2026**
WALL RESEARCH AWARD (DECLINED)	**$25 000 (JUNE 2025)**
NORTHERN SCIENTIFIC TRAINING PROGRAM	**$3300 (MAY 2025)**
NSERC CANADA GRADUATE SCHOLARSHIP – MASTER'S (CGS-M)	**$27 000 (APR. 2025)**
WCS WESTON FAMILY BOREAL RESEARCH FELLOWSHIP	**$10 000 (MAR. 2025)**
BRITISH COLUMBIA GRADUATE SCHOLARSHIP	**$17 500 (MAY 2024)**
NSERC UNDERGRADUATE STUDENT RESEARCH AWARD (USRA)	**$6000 (MAY 2024)**
UBC GEOGRAPHY ALUMNI SCHOLARSHIP	**$3450 (NOV. 2023)**

Memberships & Extracurriculars

ASLO STUDENT MEMBER	**2025/2026**
CGU STUDENT MEMBER	**2025/2026**